Inverse Kinematics Notes

Overview

These notes document the leg mathematics used for the hexapod design.

Joint definitions

- **J1 (coxa)** rotates around the z**-axis**.
- **J2 (femur)** rotates around the x**-axis**.
- **J3 (tibia)** rotates relative to the femur and determines the bend of the leg in the reduced 2D plane.

Link lengths

$$\begin{aligned} j2L &= 80\,\text{mm} \quad \text{(distance from femur joint J2 to tibia joint J3)} \\ j3L &= 120\,\text{mm} \quad \text{(distance from tibia joint J3 to foot tip)} \end{aligned}$$

Reduced 2D geometry

After the coxa orientation is accounted for, the femur and tibia are solved in a 2D plane using the coordinates:

- y: horizontal distance in the reduced leg plane
- z: vertical distance in the reduced leg plane

The straight-line distance from J2 to the foot tip is:

$$L^2 = z^2 + y^2$$

or equivalently

$$L = \sqrt{z^2 + y^2}$$

Triangle solution

The triangle formed by the femur, tibia, and the J2-to-tip distance has side lengths:

$$\begin{aligned} j2L &= 80\,\text{mm} \\ j3L &= 120\,\text{mm} \\ L &= \text{distance from J2 to the foot tip} \end{aligned}$$

Tibia angle

Using the law of cosines, the tibia angle is:

$$j3 \text{ angle} = \arccos\left(\frac{j2L^2 + j3L^2 - L^2}{2\, j2L\, j3L}\right)$$

Auxiliary angle B

Also by the law of cosines:

$$B \text{ angle} = \arccos\left(\frac{j2L^2 + L^2 - j3L^2}{2\, j2L\, L}\right)$$

Auxiliary angle A

The angle from the horizontal to the J2-to-tip line is:

$$A \text{ angle} = \arctan\left(\frac{z}{y}\right)$$

Femur angle

The femur joint angle is:

$$j2 \text{ angle} = B \text{ angle} - A \text{ angle}$$

Servo-side angle conventions

The geometric angles above are converted to servo command angles with side-specific sign conventions.

Right side

$$\begin{aligned} \text{servo } j2 \text{ angle} &= 90^\circ - j2 \text{ angle} \\ \text{servo } j3 \text{ angle} &= 90^\circ - (c + j3 \text{ angle}) \end{aligned}$$

Left side

$$\begin{aligned} \text{servo } j2 \text{ angle} &= 90^\circ + j2 \text{ angle} \\ \text{servo } j3 \text{ angle} &= 90^\circ + (c + j3 \text{ angle}) \end{aligned}$$

Here, c is a constant calibration angle used to account for the mechanical mounting and resting orientation of the tibia servo.

Resting point

The resting reduced-plane coordinates are:

$$z = -120\,\text{mm}$$
$$y = 130\,\text{mm}$$

So the resting J2-to-tip distance is:

$$L_{\text{rest}} = \sqrt{(-120)^2 + 130^2}$$

Summary

The full process is:

1. Determine the reduced-plane coordinates (y, z) after the coxa orientation is handled.
2. Compute
$$L = \sqrt{z^2 + y^2}$$
3. Compute the tibia angle:
$$j3 \text{ angle} = \arccos\left(\frac{j2L^2 + j3L^2 - L^2}{2\, j2L\, j3L}\right)$$
4. Compute
$$B \text{ angle} = \arccos\left(\frac{j2L^2 + L^2 - j3L^2}{2\, j2L\, L}\right)$$
5. Compute
$$A \text{ angle} = \arctan\left(\frac{z}{y}\right)$$
6. Compute the femur angle:
$$j2 \text{ angle} = B \text{ angle} - A \text{ angle}$$
7. Convert the geometric angles into servo command angles using the left/right formulas above.